

NoSQL Databases

Prepared By: Dr. D. P. Singh

Overview

- **NoSQL (Not Only SQL)** databases are **non-relational databases** designed to store and manage large volumes of structured, semi-structured, or unstructured data.
- They are commonly used in modern applications because they provide:
 - High scalability
 - Fast performance
 - Flexible schema
 - Distributed storage
 - High availability
- Unlike traditional SQL databases, **NoSQL databases do not require fixed table structures.**

Structured, Semi-Structured, and Unstructured Data

1. Structured Data

Structured data is **organized in a predefined format** such as rows and columns. It **follows a fixed schema**, making it easy to store, search, and analyze.

Characteristics

- Fixed schema
- Stored in tables
- Easy to query using SQL
- High consistency
- Used in relational databases

Structured, Semi-Structured, and Unstructured Data contd...

1. Structured Data (Example)

- Student Database Table

Student_ID	Name	Age	Department	CGPA
101	X	20	CSE	8.5
102	Y	21	IT	7.9
103	z	20	ECE	9.1

- Here every row has the same columns.

Structured, Semi-Structured, and Unstructured Data contd...

1. Structured Data

- **Real-world Examples**

- Banking transactions
- Employee records
- Airline reservation systems
- Hospital patient records
- Inventory management

- **Storage**

- Typically **stored in relational database** systems such as:
 - MySQL
 - PostgreSQL
 - Oracle Database

Structured, Semi-Structured, and Unstructured Data contd...

2. Semi-Structured Data

- Semi-structured **data does not follow a rigid table structure, but it contains tags, keys, or metadata** that help organize the data.
- There is **no fixed schema**, but the data is still partially organized.
- **Characteristics**
 - Flexible schema
 - Uses key-value pairs or tags
 - Easy to modify
 - Self-describing
 - Common in NoSQL databases

Structured, Semi-Structured, and Unstructured Data

2. Semi-Structured Data (Examples)

contd...

JSON

```
{  
  "StudentID": 101,  
  "Name": "X",  
  "Department": "CSE",  
  "Skills": [  
    "Java",  
    "Python",  
    "SQL"  
  ]  
}
```

XML

```
<Student>  
  <StudentID>101</StudentID>  
  <Name>X</Name>  
  <Department>CSE</Department>  
</Student>
```

Notice that another student may have extra fields

Structured, Semi-Structured, and Unstructured Data contd...

2. Semi-Structured Data

- **Real-world Examples**

- JSON files
- XML documents
- HTML pages
- Email messages
- Log files

- **Storage**

Often stored in NoSQL databases like:

- MongoDB
- CouchDB

Structured, Semi-Structured, and Unstructured Data contd...

3. Unstructured Data

- Unstructured data **has no predefined format or schema.**
- It cannot be stored naturally in rows and columns.
- Most of the world's data is unstructured.
- **Characteristics**
 - No fixed format
 - Difficult to search directly
 - Requires AI or specialized processing
 - Large storage requirement

Structured, Semi-Structured, and Unstructured Data contd...

3. Unstructured Data

• Real-world Examples

- Images
- Videos
- Audio recordings
- PDF documents
- Word documents
- Social media posts
- CCTV footage
- Medical scans

• Storage

Often stored in systems like:

- Apache Hadoop
- Amazon S3
- Google Cloud Storage

Comparison

Feature	Structured Data	Semi-Structured Data	Unstructured Data
Schema	Fixed	Flexible	None
Organization	Rows & Columns	Tags, Keys, Metadata	No organization
Storage	Relational DBMS	NoSQL Databases	File Systems/Object Storage
Query Language	SQL	JSON/XML Queries	AI, Search Engines, NLP
Flexibility	Low	Medium	High
Processing Speed	Fast	Moderate	Slow
Examples	Student records, Banking	JSON, XML, Emails	Images, Videos, Audio

Everyday Example: Online Shopping

Data	Type	Reason
Product ID = 1001, Price = ₹999	Structured	Stored in database tables
Product details in JSON (color, size, specifications)	Semi-Structured	Flexible attributes vary by product
Product images	Unstructured	Image files have no tabular schema
Customer review text	Unstructured	Free-form text
Invoice PDF	Unstructured	Document without a fixed database schema

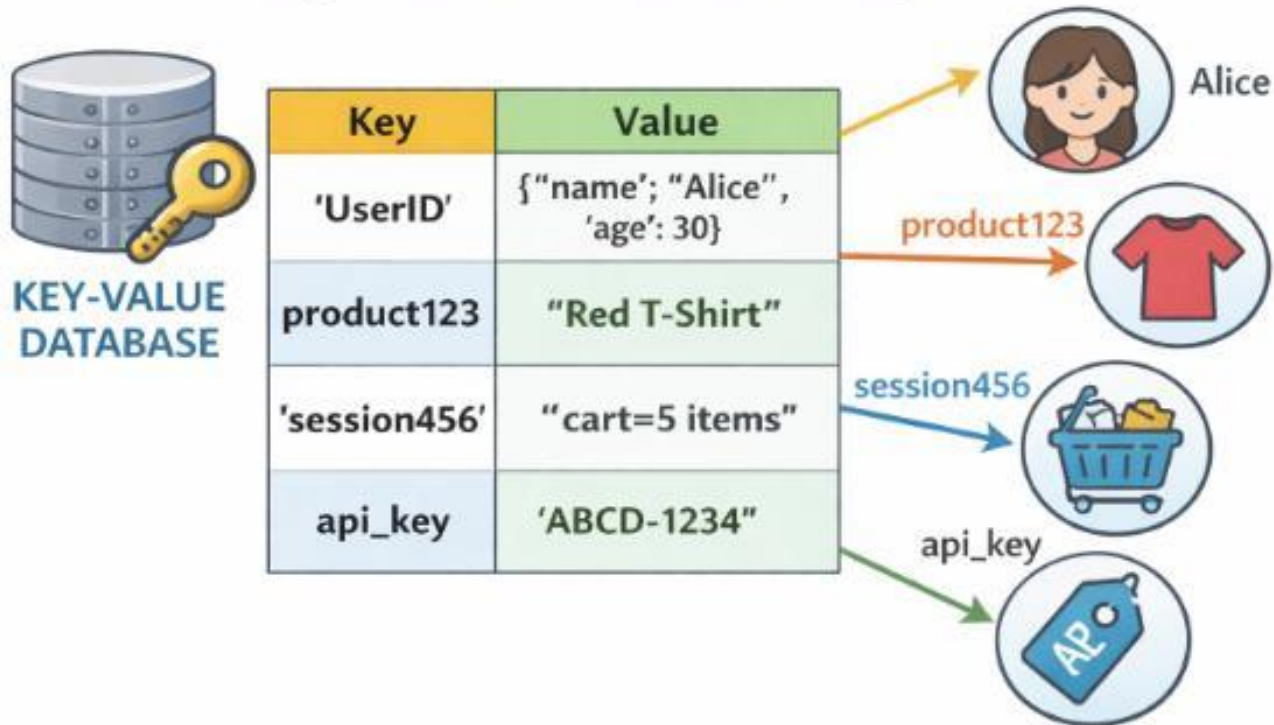
Types of NoSQL Databases

- There are **four major types** of NoSQL databases:
 - Key-Value Database
 - Document Database
 - Column-Family (Columnar) Database
 - Graph Database

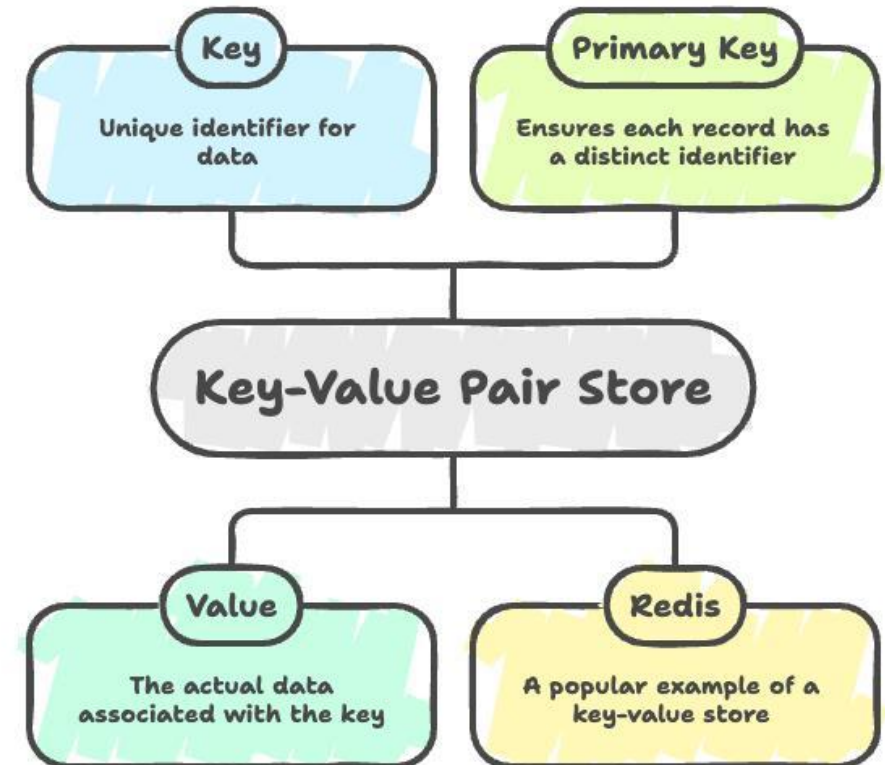
Key-Value Database

- **Key-Value Database** stores data as a collection of **keys** and their corresponding **values**.
- Think of it like a dictionary or hash map. **Key->Value**

Key-Value Database Example



Understanding Key-Value Pair Stores



Key-Value Database

- **Popular Databases**

- Redis
- Amazon DynamoDB
- Riak

- **Use Cases**

- Shopping carts
- User sessions
- Caching
- Gaming leaderboards
- Real-time recommendations

- **Advantages**

- Extremely fast lookups
- Easy to scale
- Simple data model
- Excellent performance

- **Disadvantages**

- Cannot perform complex queries
- Limited relationships between data

Document Database

- A **Document Database** stores data as **documents**, usually in JSON, BSON, or XML format.
- Each document can have different fields.



Document Database

- **Popular Databases**

- MongoDB
- CouchDB
- Firebase Firestore

- **Use Cases**

- E-commerce websites
- Content Management Systems (CMS)
- Product catalogs
- Mobile applications
- User profiles

- **Advantages**

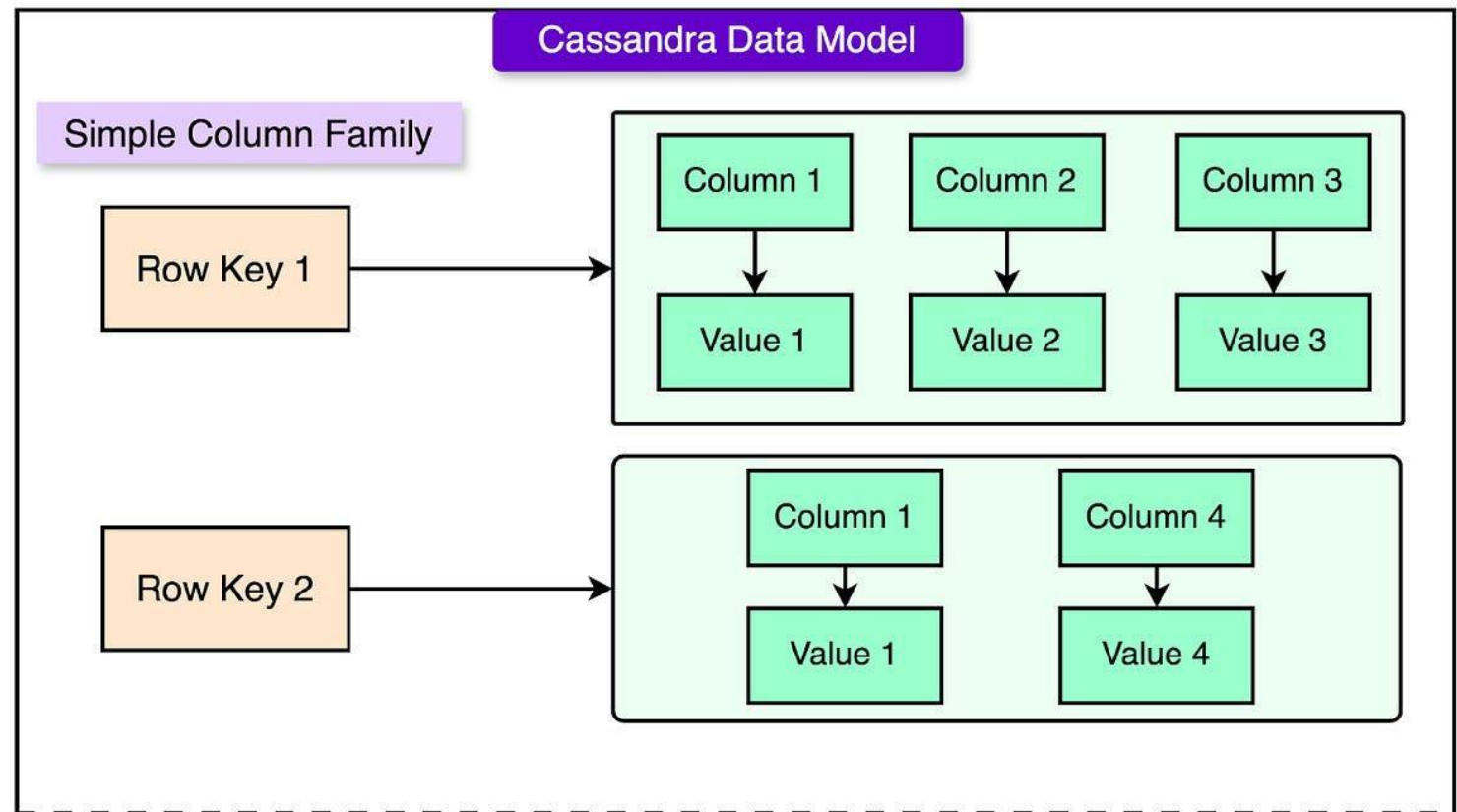
- Flexible schema
- Easy to store nested objects
- Fast development
- Good scalability

- **Disadvantages**

- Complex joins are difficult
- Data duplication may occur

Column-Family (Columnar) Database

- A **Column-Family Database** stores data in **column families** instead of traditional **rows**.
- Different rows can have different columns.



Column- Family (Columnar) Database

- **Example**

Social media analytics

User101

Likes : 150

Posts : 40

Followers : 1200

Another User: User102

Followers : 500

Posts : 25

Videos : 80

Comments : 300

Each row can contain different columns.

Column- Family (Columnar) Database

- **Popular Databases**

- Apache Cassandra
- Apache HBase
- Google Bigtable

- **Use Cases**

- Big Data
- IoT sensor data
- Banking transaction logs
- Time-series data
- Recommendation engines

- **Advantages**

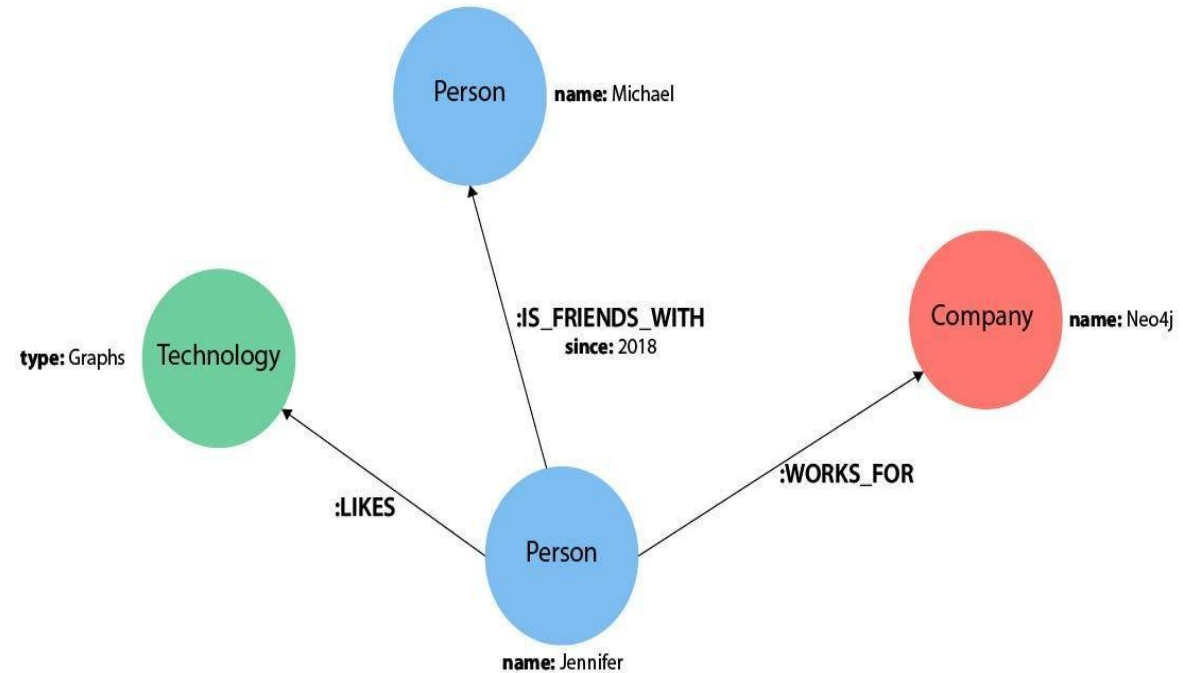
- Handles massive data volumes
- High write performance
- Horizontally scalable
- Fault tolerant

- **Disadvantages**

- Complex data modeling
- Limited support for joins

Graph Database

- A **Graph Database** stores data as **Nodes, Edges (Relationships),** and **Properties.**
- It is ideal for representing connected data.
- Nodes-> People, Products, Places
- Edges-> Relationships
- Properties-> Details about nodes



Graph Database

- **Popular Databases**

- Neo4j
- Amazon Neptune
- JanusGraph

- **Use Cases**

- Social networks
- Fraud detection
- Recommendation systems
- Knowledge graphs
- Network topology
- Route optimization

- **Advantages**

- Very fast relationship queries
- Excellent for connected data
- Flexible schema

- **Disadvantages**

- Not ideal for simple tabular data
- More complex to design and maintain

Comparison of NoSQL Databases

Type	Data Model	Best Use Cases	Popular Examples
Key-Value	Key → Value pairs	Caching, sessions, shopping carts	Redis, DynamoDB
Document	JSON/BSON/XML documents	User profiles, CMS, product catalogs	MongoDB, CouchDB
Column-Family	Rows with dynamic columns	Big Data, IoT, analytics	Cassandra, HBase
Graph	Nodes and relationships	Social networks, fraud detection, recommendations	Neo4j, Amazon Neptune

SQL vs NoSQL

Feature	SQL	NoSQL
Schema	Fixed	Flexible
Data Storage	Tables	Key-Value, Documents, Columns, Graphs
Scalability	Vertical scaling	Horizontal scaling
Relationships	Strong support with joins	Limited joins (except Graph databases)
Best For	Banking, ERP, inventory	Big Data, real-time apps, social media

Thank You